

CS 506: Introduction to Quantum Computing

Period Finding Using the Quantum Fourier Transform

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1 Overview

This lecture analyzes the quantum period-finding subroutine used in Shor's Algorithm. The goal is to determine the order r of the function

$$f(x) = a^x \pmod{N},$$

which enables efficient factorization of a composite integer N .

2 Quantum State Preparation

Let $Q = 2^m$ such that

$$N^2 \leq Q < 2N^2.$$

After initializing the quantum registers and applying Hadamard gates to the first register, we obtain the uniform superposition

$$\frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle |0\rangle.$$

Applying the modular exponentiation operator U_f gives

$$U_f |x\rangle |0\rangle = |x\rangle |a^x \pmod{N}\rangle,$$

resulting in the state

$$|\psi\rangle = \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle |a^x \pmod{N}\rangle.$$

Measuring the second register collapses the system into a periodic superposition:

$$|\phi\rangle = \frac{1}{\sqrt{\theta}} \sum_{z=0}^{\theta-1} |x_0 + zr\rangle,$$

where r is the period and

$$\theta \approx \frac{Q}{r}.$$

3 Quantum Fourier Transform

The Quantum Fourier Transform over $Q = 2^m$ basis states is defined as

$$\text{QFT}_Q |x\rangle = \frac{1}{\sqrt{Q}} \sum_{y=0}^{Q-1} e^{\frac{2\pi i xy}{Q}} |y\rangle.$$

Applying the QFT to $|\phi\rangle$ yields

$$\text{QFT}_Q |\phi\rangle = \frac{1}{\sqrt{\theta Q}} \sum_{y=0}^{Q-1} \sum_{z=0}^{\theta-1} e^{\frac{2\pi i (x_0 + zr)y}{Q}} |y\rangle.$$

Factoring the exponential term gives

$$= \frac{1}{\sqrt{\theta Q}} \sum_{y=0}^{Q-1} e^{\frac{2\pi i x_0 y}{Q}} \left(\sum_{z=0}^{\theta-1} e^{\frac{2\pi i z r y}{Q}} \right) |y\rangle.$$

The inner summation forms a geometric series that determines the measurement probabilities.

4 Measurement Probability

The probability of measuring a value $y = y_0$ is

$$P(y_0) = \frac{1}{Q\theta} \left| \sum_{z=0}^{\theta-1} e^{\frac{2\pi i z r y_0}{Q}} \right|^2.$$

Let

$$\omega = e^{\frac{2\pi i r y_0}{Q}}.$$

Then

$$P(y_0) = \frac{1}{Q\theta} \left| \sum_{z=0}^{\theta-1} \omega^z \right|^2.$$

Using the geometric series identity,

$$\sum_{z=0}^{\theta-1} \omega^z = \begin{cases} \theta, & \text{if } \omega = 1, \\ \frac{1 - \omega^\theta}{1 - \omega}, & \text{otherwise.} \end{cases}$$

4.1 Case 1: Exact Multiples of Q/r

If

$$y_0 = k \frac{Q}{r}, \quad k \in \{0, 1, \dots, r-1\},$$

then

$$\omega = e^{2\pi i k} = 1.$$

Therefore,

$$\sum_{z=0}^{\theta-1} \omega^z = \theta,$$

and

$$P(y_0) = \frac{1}{Q\theta} \theta^2 = \frac{\theta}{Q}.$$

Since $\theta = Q/r$, we obtain

$$P(y_0) = \frac{1}{r}.$$

Thus, the measurement outcomes are uniformly distributed over the multiples

$$y_k = k \frac{Q}{r}.$$

4.2 Case 2: Non-Multiples of Q/r

If y_0 is not a multiple of Q/r , then $\omega \neq 1$. In this case,

$$\sum_{z=0}^{\theta-1} \omega^z = \frac{1 - \omega^\theta}{1 - \omega}.$$

Since $\theta \approx Q/r$,

$$\omega^\theta = e^{\frac{2\pi i r y_0}{Q} \theta} = e^{2\pi i y_0} = 1.$$

Hence, the numerator vanishes and the amplitude cancels due to destructive interference:

$$\sum_{z=0}^{\theta-1} \omega^z = 0.$$

Therefore,

$$P(y_0) = 0,$$

or is negligible when Q/r is not an integer. This explains why measurements concentrate near integer multiples of Q/r .

4.3 Special Case: When $r \mid Q$

If the period divides Q , then

$$\theta = \frac{Q}{r}.$$

The measurement outcomes are

$$y_k = k \frac{Q}{r}, \quad k \in \{0, 1, \dots, r-1\},$$

uniformly at random.

Dividing by Q gives

$$\frac{y_k}{Q} = \frac{k}{r}.$$

The period r can then be recovered using the continued fractions algorithm.

Example

From the lecture,

$$\frac{y}{2^m} = \frac{15}{24} = \frac{5}{8}.$$

This yields the candidate period

$$r = 8.$$

5 Success Probability

Let k be chosen uniformly at random from $\{0, 1, \dots, r-1\}$. Then

$$\Pr(\gcd(k, r) = 1) = \Omega\left(\frac{1}{\log \log r}\right).$$

Repeating the algorithm

$$O(\log \log r)$$

times yields a constant success probability.

Since $r \leq N$ and $n = \log_2 N$, we have

$$\log r \leq n \quad \text{and} \quad \log \log r \leq \log n,$$

so the repetition complexity is

$$O(\log n).$$

6 Quantum Fourier Transform Complexity

For an n -qubit system, the Quantum Fourier Transform requires

$$O(n^2)$$

quantum gates, providing an exponential advantage in period-finding tasks.